1 I

1.1 (1)

Due to translational invariance along the z-axis, the magnetic field is independent of z. Similarly, rotational symmetry about the z-axis implies that its cylindrical components are independent of ϕ.

For an infinite straight wire carrying a steady current, the magnetic field is directed along $\hat{\phi}$, according to the right-hand rule. Thus,

$$\vec{B}(r) = B_\phi(r)\hat{\phi}.$$

Applying Ampère's law to a circular loop of radius r centered on the wire,

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I.$$

By symmetry,

$$2\pi r B_\phi(r) = \mu_0 I.$$

Therefore,

$$\boxed{\vec{B}(r) = \frac{\mu_0 I}{2\pi r}\hat{\phi}}.$$

1.2 (2)

When the current $I(t)$ increases slowly, the magnetic field can be approximated by

$$\vec{B}(r,t) = \frac{\mu_0 I(t)}{2\pi r}\hat{\phi}.$$

Consider the rectangular surface S shown in the problem, whose normal is along $\hat{\phi}$.

Let its two vertical sides be located at r_a and r_b, with $r_b > r_a$, and let their length along the z-axis be L.

Due to translational symmetry along the z-axis, the induced electric field is independent of z.

Applying Faraday's law,

$$\oint_{\partial S} \vec{E} \cdot d\vec{l} = -\frac{d}{dt}\int_S \vec{B} \cdot d\vec{S}.$$

For the circulation along the boundary, the contributions from the two radial sides cancel, while the vertical sides give

$$\oint_{\partial S} \vec{E} \cdot d\vec{l} = L[E_z(r_a) - E_z(r_b)].$$

Meanwhile, the magnetic flux through the surface is

$$\begin{aligned}\Phi_B &= \int_S \vec{B} \cdot d\vec{S} \\ &= L\int_{r_a}^{r_b} \frac{\mu_0 I(t)}{2\pi r}\, dr \\ &= \frac{\mu_0 L I(t)}{2\pi}\ln\frac{r_b}{r_a}.\end{aligned}$$

Thus,

$$L[E_z(r_a) - E_z(r_b)] = -\frac{\mu_0 L\dot{I}(t)}{2\pi} \ln \frac{r_b}{r_a}.$$

Therefore,

$$\boxed{E_z(r_a) - E_z(r_b) = -\frac{\mu_0 \dot{I}(t)}{2\pi} \ln \frac{r_b}{r_a}}.$$

For $\dot{I}(t) > 0$, the induced electric field has a more negative z-component closer to the wire.

With the causal boundary condition for a field generated by the increasing current, the induced electric field points along the negative z-direction, in accordance with Lenz's law:

$$\boxed{\vec{E}_{\text{ind}} \parallel -\hat{z}}.$$

2 II

2.1 (3)

The electric and magnetic fields are defined as

$$\vec{E} = -\frac{\partial \vec{A}}{\partial t}, \qquad \vec{B} = \nabla \times \vec{A},$$

with the Coulomb gauge condition

$$\nabla \cdot \vec{A} = 0.$$

We verify Maxwell's equations (i)–(iii).

(i)

$$\begin{aligned} \nabla \cdot \vec{E} &= -\nabla \cdot \frac{\partial \vec{A}}{\partial t} \\ &= -\frac{\partial}{\partial t}(\nabla \cdot \vec{A}) \\ &= 0. \end{aligned}$$

(ii)

$$\begin{aligned} \nabla \times \vec{E} &= -\nabla \times \frac{\partial \vec{A}}{\partial t} \\ &= -\frac{\partial}{\partial t}(\nabla \times \vec{A}) \\ &= -\frac{\partial \vec{B}}{\partial t}. \end{aligned}$$

(iii)

Using the vector identity $\nabla \cdot (\nabla \times \vec{X}) = 0$,

$$\begin{aligned} \nabla \cdot \vec{B} &= \nabla \cdot (\nabla \times \vec{A}) \\ &= 0. \end{aligned}$$

Therefore, Maxwell's equations (i)–(iii) are satisfied.

2.2 (4)

Using Maxwell's equation (iv),

$$\nabla \times \vec{B} = \mu_0 \vec{j} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}.$$

Substituting the expressions for $\vec{E}$ and $\vec{B}$ gives

$$\nabla \times (\nabla \times \vec{A}) = \mu_0 \vec{j} - \mu_0 \epsilon_0 \frac{\partial^2 \vec{A}}{\partial t^2}.$$

Using the vector identity

$$\nabla \times (\nabla \times \vec{A}) = \nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A},$$

and the Coulomb gauge condition, we obtain

$$-\nabla^2 \vec{A} = \mu_0 \vec{j} - \mu_0 \epsilon_0 \frac{\partial^2 \vec{A}}{\partial t^2}.$$

Rearranging,

$$\boxed{\left(\nabla^2 - \mu_0 \epsilon_0 \frac{\partial^2}{\partial t^2}\right) \vec{A} = -\mu_0 \vec{j}}.$$

Comparing this with the standard wave equation,

$$\left(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}\right) \vec{A} = -\mu_0 \vec{j},$$

we find

$$\boxed{c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}}.$$

3 III

3.1 (5)

The retarded solution of the wave equation is

$$\vec{A}(\vec{r}, t) = \frac{\mu_0}{4\pi} \int \frac{\vec{j}(\vec{r}', t - |\vec{r} - \vec{r}'|/c)}{|\vec{r} - \vec{r}'|} \, d^3 r'.$$

For the current

$$\vec{j}(\vec{r}, t) = I \Theta(t) \delta(x) \delta(y) \hat{z},$$

the delta functions restrict the integral to the z-axis.

At a point $(x, y, 0)$, we have

$$|\vec{r} - \vec{r}'| = \sqrt{r^2 + z'^2}, \qquad r = \sqrt{x^2 + y^2}.$$

Therefore,

$$\vec{A}(r, t) = \frac{\mu_0 I}{4\pi} \hat{z} \int_{-\infty}^{\infty} \frac{\Theta\left(t - \frac{\sqrt{r^2 + z'^2}}{c}\right)}{\sqrt{r^2 + z'^2}} \, dz'.$$

(a) $r > ct$

Since

$$\sqrt{r^2 + z'^2} \geq r > ct,$$

we have

$$t - \frac{\sqrt{r^2 + z'^2}}{c} < 0.$$

Thus, the step function vanishes throughout the integration region, giving

$$\boxed{\vec{A}(r,t) = 0 \qquad (r > ct)}.$$

(b) $0 < r < ct$

The step function is nonzero when

$$t - \frac{\sqrt{r^2 + z'^2}}{c} > 0,$$

or equivalently,

$$r^2 + z'^2 < (ct)^2.$$

Hence, the integration limits become

$$-\sqrt{(ct)^2 - r^2} < z' < \sqrt{(ct)^2 - r^2}.$$

Therefore,

$$\vec{A}(r,t) = \frac{\mu_0 I}{4\pi}\hat{z}\int_{-\sqrt{(ct)^2-r^2}}^{\sqrt{(ct)^2-r^2}} \frac{dz'}{\sqrt{r^2 + z'^2}}.$$

Using the integral formula

$$\int \frac{dz'}{\sqrt{r^2 + z'^2}} = \ln\left(z' + \sqrt{r^2 + z'^2}\right) + C,$$

we obtain

$$\vec{A}(r,t) = \frac{\mu_0 I}{4\pi} \ln\left(\frac{ct + \sqrt{(ct)^2 - r^2}}{ct - \sqrt{(ct)^2 - r^2}}\right)\hat{z}.$$

Introducing

$$F \equiv \sqrt{1 - \left(\frac{r}{ct}\right)^2},$$

the result can be written as

$$\boxed{\vec{A}(r,t) = \frac{\mu_0 I}{4\pi} \ln\frac{1+F}{1-F}\,\hat{z} \qquad (0 < r < ct)}.$$

3.2 (6)

The magnetic field is obtained from

$$\vec{B} = \nabla \times \vec{A}.$$

(a) $r > ct$

Since $\vec{A} = 0$ throughout this region,

$$\boxed{\vec{B}(r,t) = 0 \qquad (r > ct)}.$$

(b) $0 < r < ct$

Since

$$\vec{A}(r,t) = A_z(r,t)\hat{z},$$

the curl in cylindrical coordinates becomes

$$\vec{B} = -\frac{\partial A_z}{\partial r}\hat{\phi}.$$

From (5),

$$A_z(r,t) = \frac{\mu_0 I}{4\pi}\ln\frac{1+F}{1-F}.$$

Differentiating with respect to r,

$$\frac{\partial A_z}{\partial r} = \frac{\mu_0 I}{4\pi}\frac{2}{1-F^2}\frac{\partial F}{\partial r}.$$

Using

$$F = \sqrt{1 - \frac{r^2}{(ct)^2}},$$

we obtain

$$\frac{\partial F}{\partial r} = -\frac{r}{(ct)^2 F}, \qquad 1 - F^2 = \frac{r^2}{(ct)^2}.$$

Thus,

$$\begin{aligned}\frac{\partial A_z}{\partial r} &= \frac{\mu_0 I}{4\pi}\frac{2(ct)^2}{r^2}\left(-\frac{r}{(ct)^2 F}\right) \\ &= -\frac{\mu_0 I}{2\pi r F}.\end{aligned}$$

Therefore,

$$\boxed{\vec{B}(r,t) = \frac{\mu_0 I}{2\pi r F}\hat{\phi} \qquad (0 < r < ct)}.$$

The magnetic field points along the positive $\hat{\phi}$-direction.

3.3 (7)

For a fixed distance r, after a sufficiently long time has elapsed,

$$t \to \infty, \qquad \frac{r}{ct} \to 0.$$

Therefore,

$$F = \sqrt{1 - \left(\frac{r}{ct}\right)^2} \longrightarrow 1.$$

Using the result from (6),

$$\begin{aligned}\lim_{t\to\infty}\vec{B}(r,t) &= \lim_{t\to\infty}\frac{\mu_0 I}{2\pi r F}\hat{\phi} \\ &= \frac{\mu_0 I}{2\pi r}\hat{\phi}.\end{aligned}$$

Thus,

$$\boxed{\lim_{t\to\infty}\vec{B}(r,t) = \frac{\mu_0 I}{2\pi r}\hat{\phi}}.$$

This agrees with the static magnetic field obtained in (1).